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COSC 461
Maniccam

A. Numbers of epochs
Middle layers: 1. Size: 100

Epochs	Accuracy	Time(seconds)
2	0.8606	14
5	0.8744	42
7	0.8751	46
10	0.8853	71
12	0.8823	83

Summary: Increasing the number of epochs increases the runtime and it also seems to increase the accuracy to a certain degree with its peak at ten iterations, then lightly losing accuracy at twelve. The explanation for this happening would be overfitting. Due to the higher number of iterations, the network starts to memorize the training data instead of learning from it, resulting in a lower accuracy rate.

B. Number of hidden/middle layers
Epochs: 5. Size: 100

Middle layers	Accuracy	Time(seconds)
1	0.8744	41
3	0.8665	42
5	0.8529	48
7	0.8701	52
10	0.8630	58

Summary: Increasing the number of middle layers increased the runtime while decreasing the accuracy of the neural network. This is another case of overfitting, as more layers means more information for the network to run through which may result in problems such as a lower accuracy rate.

C. Size of hidden layers

Epochs: 5. Middle layers: 1

Size	Accuracy	Time(seconds)
25	0.8620	27
50	0.8640	29
75	0.8724	34
100	0.8744	41
200	0.8732	50

Summary: Increasing the size of the hidden layers resulted in an increase in runtime and an increase in accuracy up to a certain point. The accuracy peaked at a size of one hundred and decreased ever so slightly with a size of two hundred, with overfitting being the cause once again. The accuracy increase happens as the larger sizes allow for more ways to represent the data.